

Bulk translation QA

Did the population analysis work, and what requires attention?

Accepted flightlines 12	Sites 3	Observation rows 96,016,835	Excluded candidates 0	Candidate coefficients 54	Attention flags 27
Median R ² 0.996	Worst R ² 0.580	Median RMSE 20.436	Worst RMSE 267.020	Median correction 7.5%	Maximum correction 114.3%

Priority review

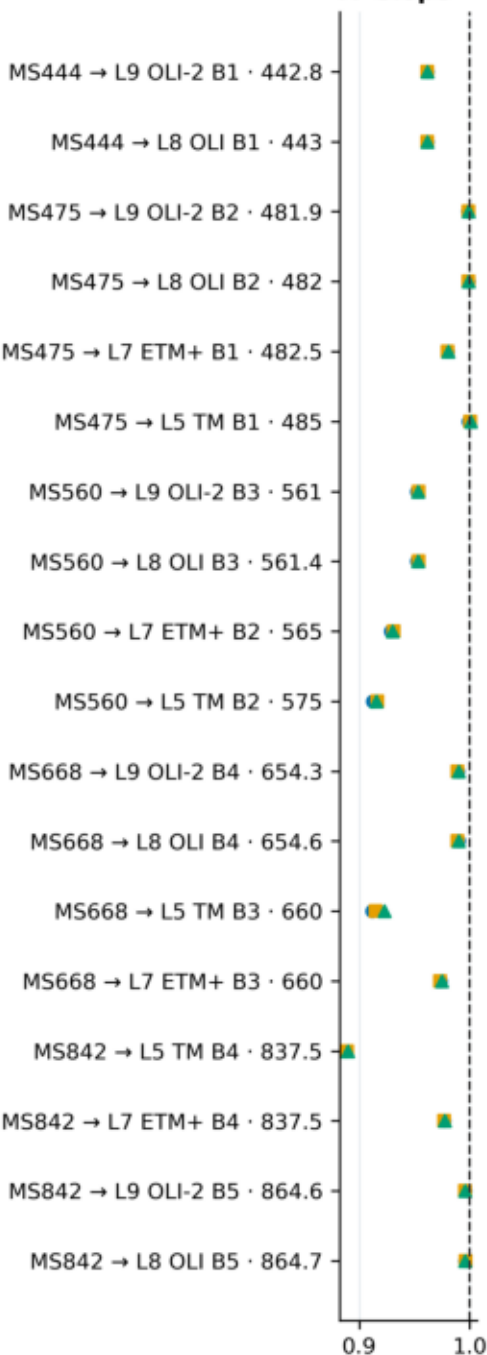
Most weighting-sensitive	MS668 → L5 TM B3 • 660	0.011
Most site-dependent	MS668 → L5 TM B3 • 660	0.698
Weakest LOSO	MS668 → L5 TM B3 • 660 • WREF	0.188

ATTENTION REQUIRED

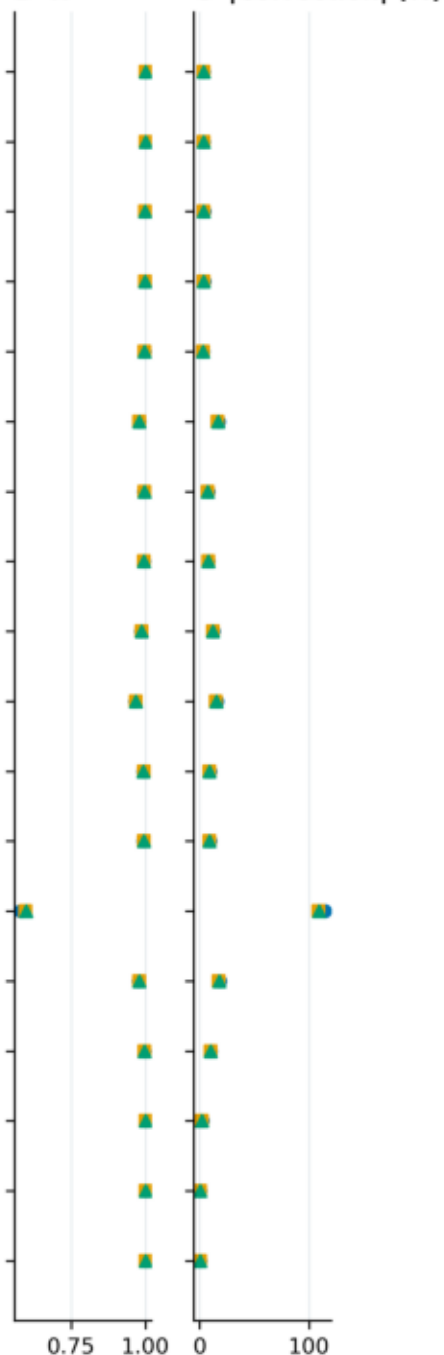
High R² alone does not establish sensor interchangeability. Review weighting, site dependence, correction magnitude, and held-out-site performance.

Translation performance

A Slope



B R²



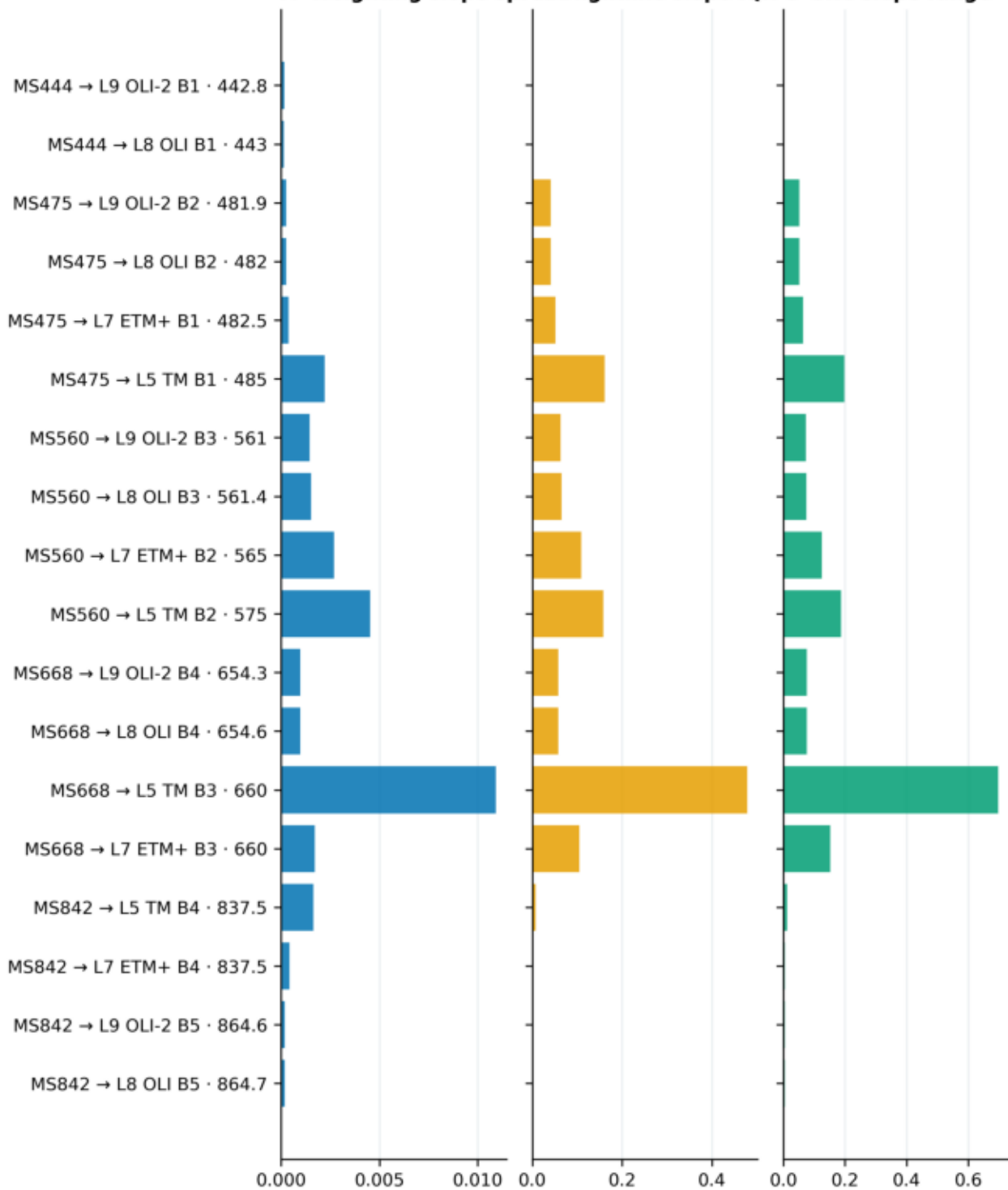
C |Correction| (%)



● pixel pooled ■ flightline balanced ▲ site balanced

Translation stability

A Weighting slope spread B Flightline slope IQR C Site slope range

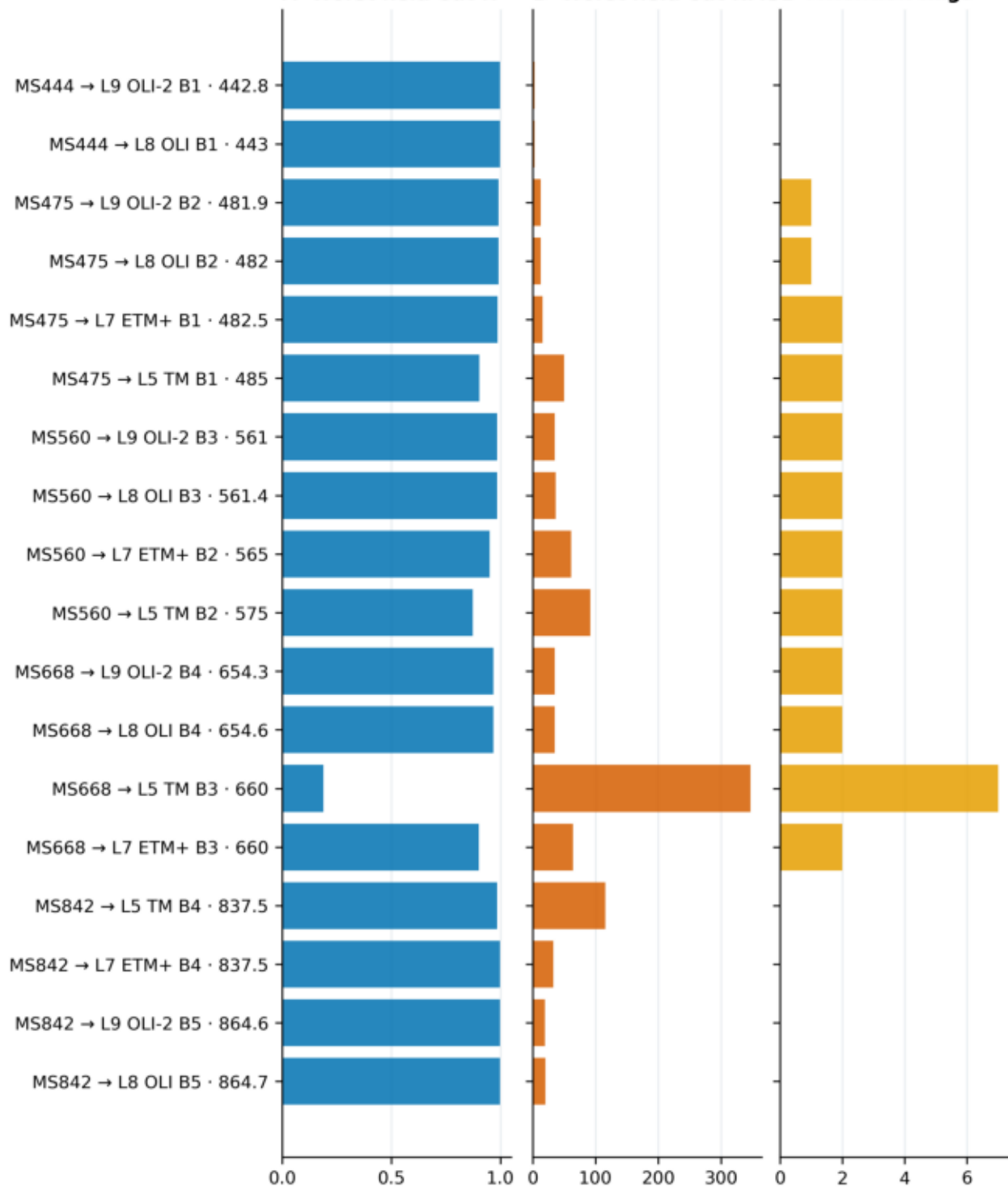


Generalization and failure cases

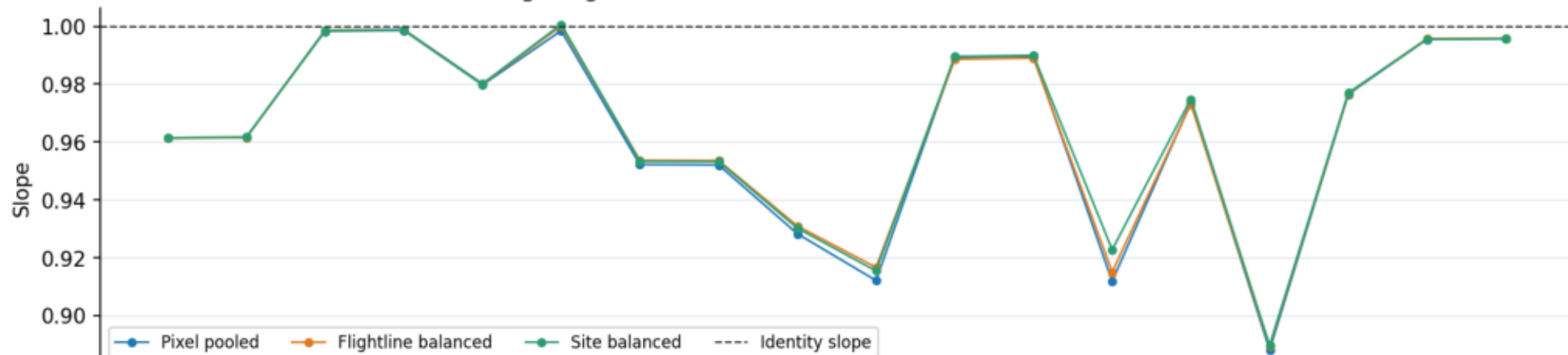
A Worst held-out R^2

B Worst held-out RMSE

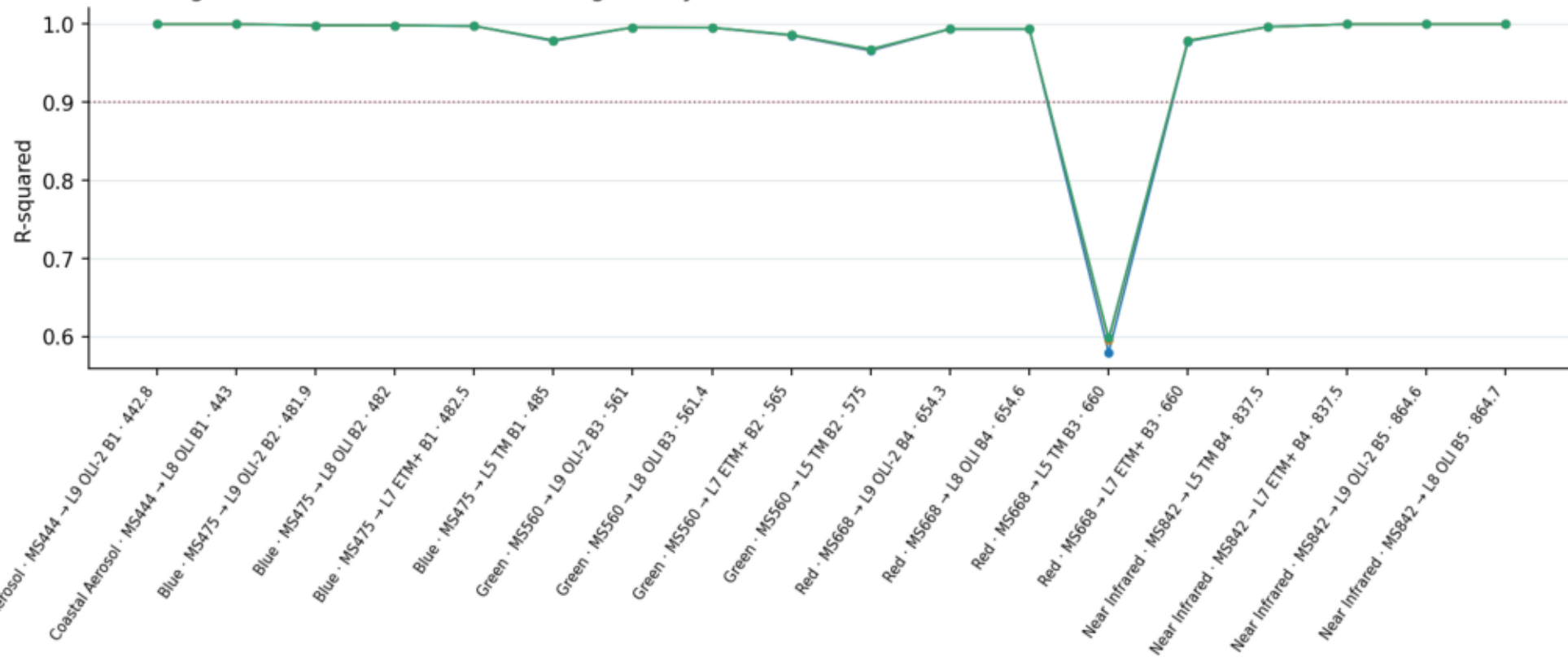
Attention flags



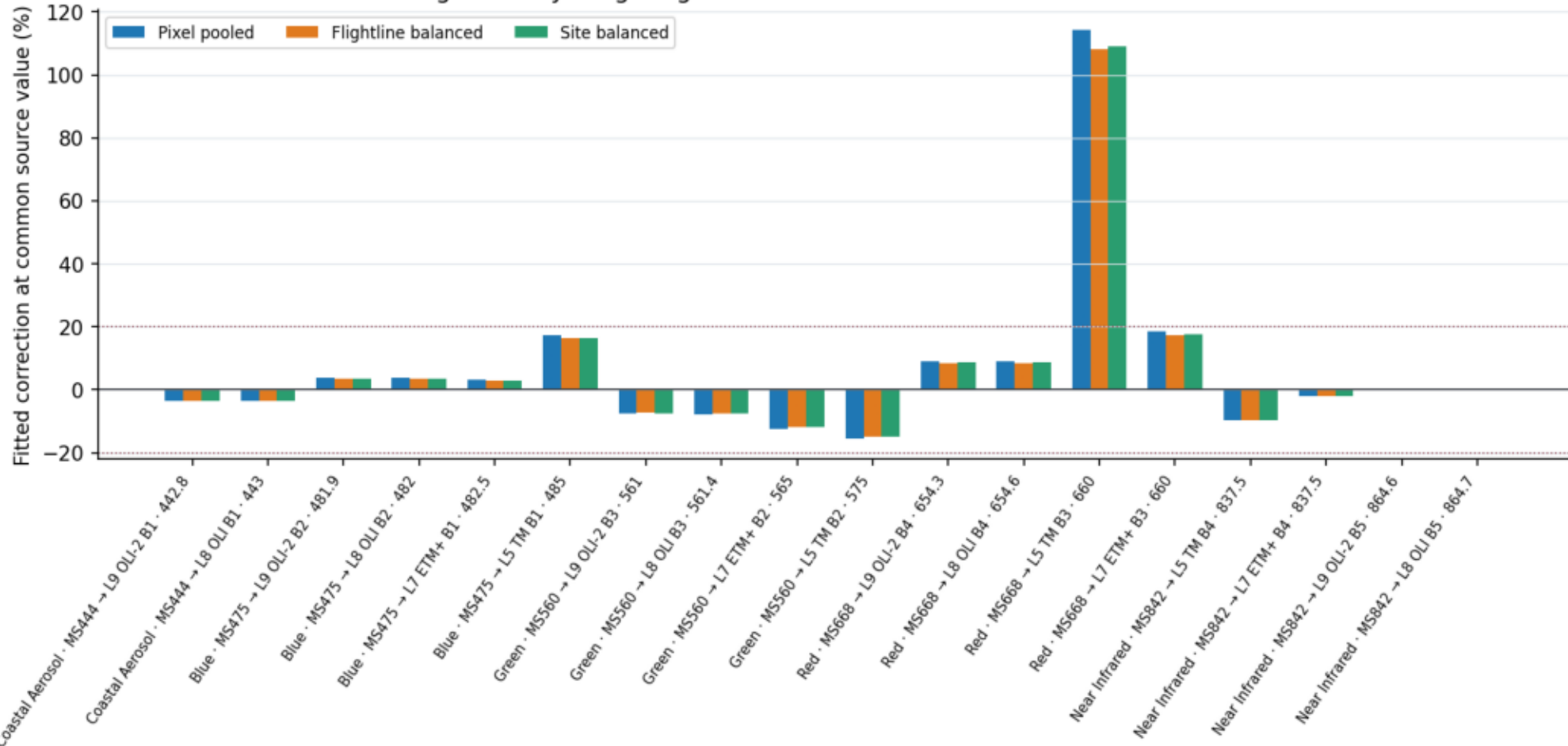
Translation coefficients across weighting choices



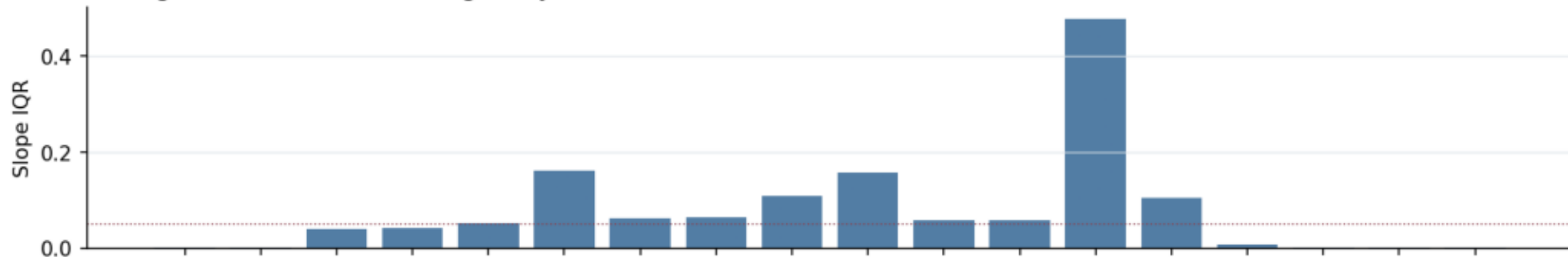
Fit strength does not establish interchangeability



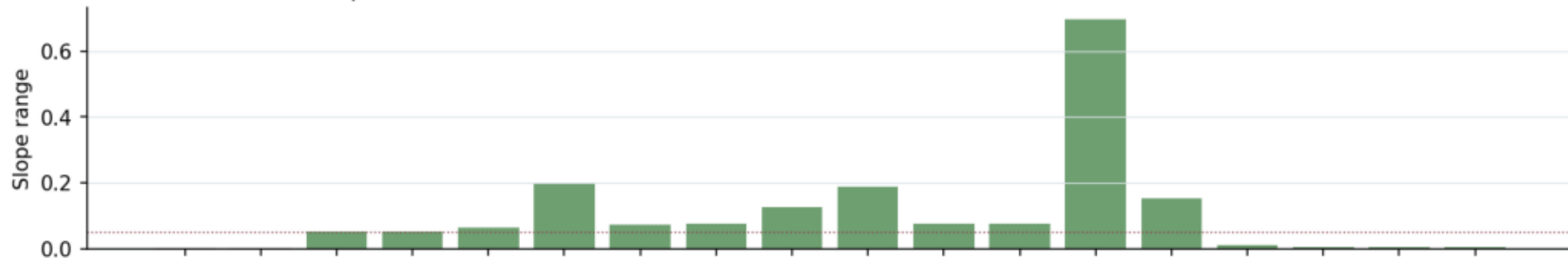
Correction direction and magnitude by weighting



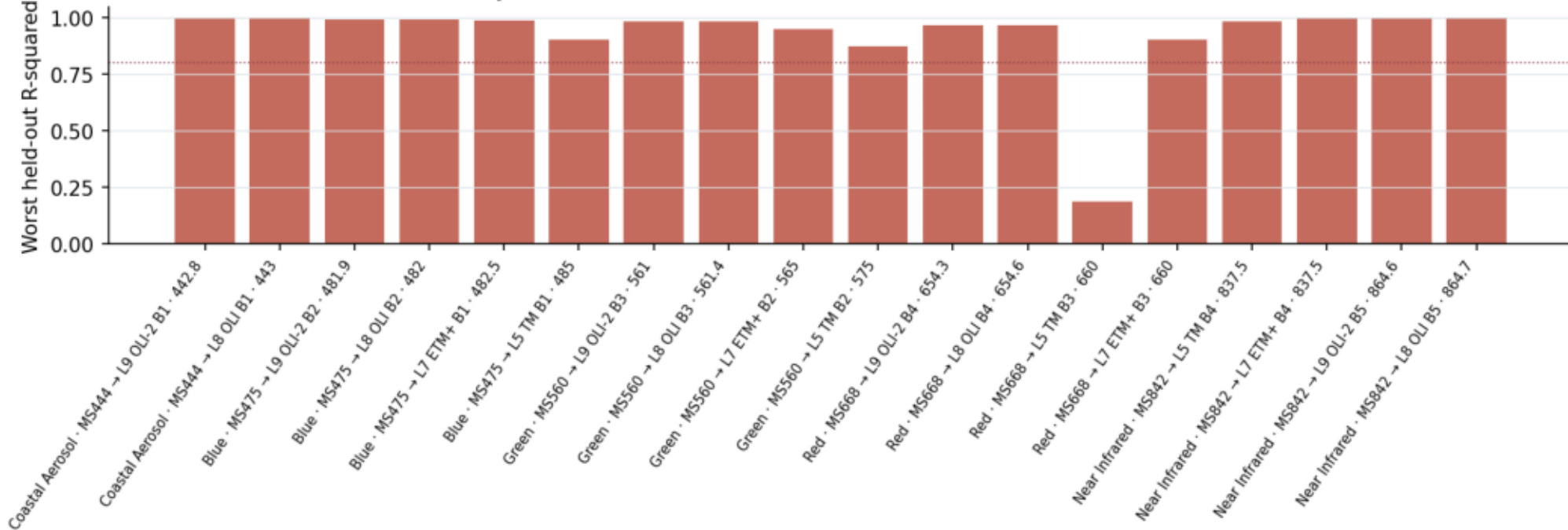
Per-flightline coefficient heterogeneity



Per-site coefficient dependence



Leave-one-site-out transferability



Ranked attention and provenance

MS B2 → L8 OLI B2: site_dependence (per_site)

MS B3 → L8 OLI B3: flightline_heterogeneity (per_flightline)

MS B3 → L8 OLI B3: site_dependence (per_site)

MS B4 → L8 OLI B4: flightline_heterogeneity (per_flightline)

MS B4 → L8 OLI B4: site_dependence (per_site)

MS B2 → L9 OLI-2 B2: site_dependence (per_site)

MS B3 → L9 OLI-2 B3: flightline_heterogeneity (per_flightline)

Machine-readable overview:

```
MS B3 → L9 OLI-2 B3: site_dependence (per_site)
  "absolute_fitted_correction_percent_max": 114.27389163542519,
  "absolute_fitted_correction_percent_median": 7.517932068066687,
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  "candidate_coefficient_count": 54,
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  "candidate_r2_min": 0.999854640565622,
  "candidate_rmse_max": 267.0201429294457,
  "candidate_rmse_median": 20.43582401718967,
  "candidate_rmse_min": 2.385772098661195,
  "candidate_slope_max": 1.0007059025745395,
  "candidate_slope_median": 0.9757285792116724,
  "candidate_slope_min": 0.8869328241274,
  "excluded_candidate_count": 0,
  "failed_or_excluded_source_count": 0,
  "leave_one_site_out_evaluation_count": 54,
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  "value": 0.6978944834307463
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  "band_index": 3,
  "label": "MS668 \u2192 L5 TM B3 \u00b7 660",
  "metric": "candidate_slope_spread",
  "value": 0.01091962498477339
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{
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  "per_flightline_fit_count": 216,
  "per_site_fit_count": 54,
  "selected_observation_rows": 96016835,
  "sites": [
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    "WREF",
    "YELL"
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  "successful_candidate_coefficient_count": 54,
  "worst_loso": {
    "band_index": 3,
    "held_out_site": "WREF",
    "label": "MS668 \u2192 L5 TM B3 \u00b7 660 \u00b7 WREF",
    "metric": "held_out_r2",
    "rmse": 347.5367561449956,
    "translation_pair": "MicaSense_to-match_TM_and_ETM+_to_Landsat_5_TM",
    "value": 0.18804518884781196
  }
}
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